Uninstalling OpenShift Data Foundation in Internal mode

Updated January 19 2026 at 10:31 PM - English

THIS PROCEDURE WILL REMOVE ALL DATA STORED IN OPENSHIFT DATA FOUNDATIONS

IMPORTANT
The following article is not made to remove only certain components of ODF, but to uninstall all of ODF

Use the steps in this section to uninstall OpenShift Data Foundation.

Uninstall Annotations

Annotations on the Storage Cluster are used to change the behavior of the uninstall process. To define the uninstall behavior, the following two annotations have been introduced in the storage cluster:

- `uninstall.ocs.openshift.io/cleanup-policy: delete`
- `uninstall.ocs.openshift.io/mode: graceful`

These annotations should be fully understood before applying them to an ocs cluster. The below table provides information on the different values that can be used with these annotations:

Uninstall annotations descriptions

```
cleanup-policy = delete
```

Default value. Rook cleans up the physical drives and the DataDirHostPath

```
cleanup-policy = retain
```

Rook does not clean up the physical drives and the DataDirHostPath

```
mode=graceful
```

Rook and NooBaa **pauses** the uninstall process until the administrator/user removes the Persistent Volume Claims (PVCs) and Object Bucket Claims (OBCs)

```
mode=forced
```

Rook and NooBaa proceed with uninstalling even if the PVCs/OBCs provisioned using Rook and NooBaa exist respectively.

Edit the value of the annotation to change the cleanup policy or the uninstall mode.

```
$ oc annotate storagecluster -n openshift-storage ocs-storagecluster
uninstall.ocs.openshift.io/cleanup-policy="<CLEANUP_POLICY>" --overwrite
```

```
$  oc annotate storagecluster -n openshift-storage ocs-storagecluster
uninstall.ocs.openshift.io/mode="<MODE>" --overwrite
```

The expected output for both commands:

```
storagecluster.ocs.openshift.io/ocs-storagecluster annotated
```

Uninstall OpenShift Data Foundation

Prerequisites

- Ensure that the OpenShift Data Foundation cluster is in a healthy state.
  The uninstall process can fail when some of the pods are not terminated successfully due to insufficient resources or nodes.
  In case the cluster is in an unhealthy state, contact Red Hat Customer Support before uninstalling OpenShift Data Foundation.
- Ensure that applications are not consuming persistent volume claims (PVCs) or object bucket claims (OBCs) using the storage classes provided by OpenShift Data Foundation.
  If any custom resources (such as custom storage classes, cephblockpools) were created by

the admin, they must be deleted by the admin after removing the resources which consumed them.

Procedure

1. Delete the volume snapshots that are using OpenShift Data Foundation.

   i. List the volume snapshots from all the namespaces.

```
$ oc get volumesnapshot --all-namespaces
```

   ii. From the output of the previous command, identify and delete the volume snapshots that are using OpenShift Data Foundation.

```
$ oc delete volumesnapshot <VOLUME-SNAPSHOT-NAME> -n <NAMESPACE>
```

`<VOLUME-SNAPSHOT-NAME>` Is the name of the volume snapshot
`<NAMESPACE>` Is the project namespace

2. Delete PVCs and OBCs that are using OpenShift Data Foundation. In the default uninstall mode (graceful), the uninstaller waits till all the PVCs and OBCs that use OpenShift Data Foundation are
deleted. If you want to delete the Storage Cluster without deleting the PVCs beforehand, you may set the uninstall mode annotation to `forced` and skip this step. Doing this results in orphan PVCs and OBCs in the system.

   i. Delete OpenShift Container Platform monitoring stack PVCs using OpenShift Data Foundation.
   For more information, see, Removing monitoring stack from OpenShift Data Foundation.
   ii. Delete OpenShift Container Platform Registry PVCs using OpenShift Data Foundation.
   For more information, see, Removing OpenShift Container Platform registry from OpenShift Data Foundation.
   iii. Delete OpenShift Container Platform logging PVCs using OpenShift Data Foundation.
   For more information, see, Removing the cluster logging operator from OpenShift Data Foundation.
   iv. Delete other PVCs and OBCs provisioned using OpenShift Data Foundation.

   - The following script is a sample script to identify the PVCs and OBCs provisioned using OpenShift Data Foundation. The script ignores the PVCs that are used internally by Openshift Data Foundation.

```
#!/bin/bash
RBD_PROVISIONER="openshift-storage.rbd.csi.ceph.com"
CEPHFS_PROVISIONER="openshift-storage.cephfs.csi.ceph.com"
NOOBAA_PROVISIONER="openshift-storage.noobaa.io/obc"
RGW_PROVISIONER="openshift-storage.ceph.rook.io/bucket"
NOOBAA_DB_PVC="noobaa-db"
NOOBAA_BACKINGSTORE_PVC="noobaa-default-backing-store-noobaa-pvc"
#Find all the OCS StorageClasses
OCS_STORAGECLASSES=$(oc get storageclasses | grep -e "$RBD_PROVISIONER" -e
"$CEPHFS_PROVISIONER" -e         "$NOOBAA_PROVISIONER" -e
"$RGW_PROVISIONER" | awk '{print $1}')
# List PVCs in each of the StorageClasses
for SC in $OCS_STORAGECLASSES
 do
   echo
"======================================================================"
   echo "$SC StorageClass PVCs and OBCs"
   echo
"======================================================================"
   oc get pvc  --all-namespaces --no-headers 2>/dev/null | grep $SC | grep
-v -e "$NOOBAA_DB_PVC" -e "  $NOOBAA_BACKINGSTORE_PVC"
   oc get obc  --all-namespaces --no-headers 2>/dev/null | grep $SC
   echo
done
```

Note: Omit `RGW_PROVISIONER` for cloud platforms.

- Delete the OBCs.

```
$ oc delete obc <obc name> -n <project name>
```

`<obc-name>` Is the name of the OBC
`<project-name>` Is the name of the project

- Delete the PVCs.

```
$ oc delete pvc <pvc name> -n <project-name>
```

`<pvc-name>` Is the name of the PVC
`<project-name>` Is the name of the project

Note: Ensure that you have removed any custom backing stores, bucket classes, etc., created in the cluster.

3. Delete the Storage Cluster object and wait for the removal of the associated resources:

```
For ODF 4.18 and lower version:
$ oc delete -n openshift-storage storagesystem --all --wait=true

For ODF 4.19 and higher version:
$ oc delete -n openshift-storage storagecluster --all --wait=true
```

IMPORTANT

- If the resources are not deleted within 5 minutes, describe the object and determine the failure through the events or status:

  ```
  $ oc describe -n openshift-storage <type> <name>
  ```

  `<type>` Specify the type of the resource. It can be an API resource or object.
  `<name>` Specify the name of the resource specified.

  Example:

  ```
  $ oc describe -n openshift-storage storagecluster ocs-storagecluster
  ```

- Determine the required resolution for the failure and implement it.

4. Check for cleanup pods if the `uninstall.ocs.openshift.io/cleanup-policy` was set to `delete` (default) and ensure that their status is `Completed` .

```
$ oc get pods -n openshift-storage | grep -i cleanup
NAME                                READY   STATUS      RESTARTS   AGE
cluster-cleanup-job-<xx>         0/1     Completed   0          8m35s
cluster-cleanup-job-<yy>             0/1     Completed   0          8m35s
cluster-cleanup-job-<zz>             0/1     Completed   0          8m35s
```

Note: If the cluster-cleanup-jobs are not created, then manually delete all the files under dataDirHostPath (default /var/lib/rook) on each storage node in the cluster where configuration is stored by ceph daemons.

Note: For ODF 4.18 and higher versions, ceph replicates the OSD metadata across different locations on the OSD disk. Make sure that all the OSD metadata is deleted from the disk, else the reinstall of ODF will fail on those disks. Follow this document to cleanup the OSD disks

5. Confirm that the directory `/var/lib/rook` is now empty. This directory will be empty only if the `uninstall.ocs.openshift.io/cleanup-policy` annotation was set to delete(default).

```
$ for i in $(oc get node -l cluster.ocs.openshift.io/openshift-storage= -o
name); do oc debug ${i} -- chroot /host ls -l /var/lib/rook; done
```

If the directory `/var/lib/rook` is not empty, remove the `/var/lib/rook`.

```
$ for i in $(oc get node -l cluster.ocs.openshift.io/openshift-storage= -o
name); do oc debug ${i} -- chroot /host rm -rf /var/lib/rook; done
```

6. If encryption was enabled at the time of install, remove `dm-crypt` managed device-mapper mapping from OSD devices on all the OpenShift Data Foundation nodes.

   i. Create a debug pod and chroot to the host on the storage node.

   ```
   $ oc debug node/<node name>
   $ chroot /host
   ```

   `<node-name>` Is the name of the node

   ii. Get Device names and make note of the OpenShift Container Storage devices.

   ```
   $ dmsetup ls
   ocs-deviceset-0-data-0-57snx-block-dmcrypt (253:1)
   ```

   iii. Remove the mapped device.

   ```
   $ cryptsetup luksClose --debug --verbose ocs-deviceset-0-data-
   0-57snx-block-dmcrypt
   ```

   Note: If the above command gets stuck due to insufficient privileges, run the following commands:

   i. Press CTRL+Z to exit the above command.
   ii. Find PID of the process which was stuck.

   ```
   $ ps -ef | grep crypt
   ```

   iii. Terminate the process using the kill command.

   ```
   $ kill -9 <PID>
   ```

   `<PID>` Is the process ID

iv. Verify that the device name is removed.

```
$ dmsetup ls
```

7. Only applicable for ODF 4.19.7 and above versions:

```
$ oc delete cm ocs-client-operator-config -n openshift-storage
```

8. Delete the namespace and wait till the deletion is complete. You need to switch to another project if openshift-storage is the active project.

   For example:

```
$ oc project default
$ oc delete project openshift-storage --wait=true --timeout=5m
```

   The project is deleted if the following command returns a `NotFound` error.

```
$ oc get project openshift-storage
```

   Note: While uninstalling OpenShift Data Foundation, if the namespace is not deleted completely and remains in Terminating state, perform the steps in Troubleshooting and deleting remaining resources during Uninstall to identify objects that are blocking the namespace from being terminated.

   Note: Make sure to remove the disaster-protection finalizers in mon and mon-endpoints after this step. For more information, see Removing the disaster-protection finalizers in mon and mon endpoints.

9. Delete the local storage operator configurations if you have deployed OpenShift Data Foundation using local storage devices. See Removing local storage operator configurations.

10. Unlabel the storage nodes.

```
$ oc label nodes  --all cluster.ocs.openshift.io/openshift-storage-
$ oc label nodes  --all topology.rook.io/rack-
```

11. Remove the OpenShift Data Foundation taint if the nodes were tainted.

```
$ oc adm taint nodes --all node.ocs.openshift.io/storage-
```

12. Confirm all PVs provisioned using OpenShift Data Foundation are deleted. If there is any PV left in the Released state, delete it.

```
$ oc get pv
$ oc delete pv _<pv-name>_
```

`<pv-name>` Is the name of the PV

13. Delete the Multicloud Object Gateway storageclass.

```
$ oc delete storageclass openshift-storage.noobaa.io --wait=true --timeout=5m
```

14. Remove `CustomResourceDefinitions` .

```
oc delete crd backingstores.noobaa.io bucketclasses.noobaa.io
cephblockpools.ceph.rook.io cephclusters.ceph.rook.io
cephfilesystems.ceph.rook.io cephnfses.ceph.rook.io
cephobjectstores.ceph.rook.io cephobjectstoreusers.ceph.rook.io
noobaas.noobaa.io ocsinitializations.ocs.openshift.io
storageclusters.ocs.openshift.io cephclients.ceph.rook.io
cephobjectrealms.ceph.rook.io cephobjectzonegroups.ceph.rook.io
cephobjectzones.ceph.rook.io cephrbdmirrors.ceph.rook.io
storagesystems.odf.openshift.io cephblockpoolradosnamespaces.ceph.rook.io
cephbucketnotifications.ceph.rook.io cephbuckettopics.ceph.rook.io
cephcosidrivers.ceph.rook.io cephfilesystemmirrors.ceph.rook.io
cephfilesystemsubvolumegroups.ceph.rook.io
csiaddonsnodes.csiaddons.openshift.io networkfences.csiaddons.openshift.io
reclaimspacecronjobs.csiaddons.openshift.io
reclaimspacejobs.csiaddons.openshift.io storageclassrequests.ocs.openshift.io
storageconsumers.ocs.openshift.io storageprofiles.ocs.openshift.io
volumereplicationclasses.replication.storage.openshift.io
volumereplications.replication.storage.openshift.io --wait=true --timeout=5m
```

15. **For ODF v4.19.0-7 clusters only:** The storageclient resource does not have a namespace defined in the yaml. Therefore, when the project is deleted, this resource can be missed. This resource needs to be deleted and will block reinstallation if not removed. If you've arrived at this step, the resource has likely been marked for deletion and just needs a finalizer patch.

```
$ oc patch storageclient ocs-storagecluster --type merge -p '{"metadata":
{"finalizers": []}}'
```

16. (Optional) If cluster-wide encryption is enabled using the kubernetes authentication method for Key Management System (KMS), you need to remove the `clusterrolebinding` .

```
$ oc delete clusterrolebinding vault-tokenreview-binding
```

17. (Optional) To ensure that the vault keys are deleted permanently, you need to manually delete the metadata associated with the vault key.

IMPORTANT
Execute this step only if Vault Key/Value (KV) secret engine API, version 2 is used for cluster-wide encryption with KMS since the vault keys are marked as deleted and not permanently deleted during the uninstallation of OpenShift Container Storage. You can always restore it later if required.

i. List the keys in the vault.

```
$ vault kv list <backend_path>
```

`<backend_path>` Is the path in the vault where the encryption keys are stored.
For example:

```
$ vault kv list kv-v2
```

Example output:

```
 Keys
 -----
NOOBAA_ROOT_SECRET_PATH/
rook-ceph-osd-encryption-key-ocs-deviceset-thin-0-data-0m27q8
rook-ceph-osd-encryption-key-ocs-deviceset-thin-1-data-0sq227
rook-ceph-osd-encryption-key-ocs-deviceset-thin-2-data-0xzszb
```

ii. List the metadata associated with the vault key.

```
$ vault kv get kv-v2/<key>
```

For the Multicloud Object Gateway (MCG) key:

```
$ vault kv get kv-v2/NOOBAA_ROOT_SECRET_PATH/<key>
```

`<key>` Is the encryption key.

For Example:

```
$ vault kv get kv-v2/rook-ceph-osd-encryption-key-ocs-deviceset-thin-0-data-0m27q8
```

Example output:

```
====== Metadata ======
Key              Value
---              -----
created_time     2021-06-23T10:06:30.650103555Z
deletion_time    2021-06-23T11:46:35.045328495Z
destroyed        false
version          1
```

ii. Delete the metadata.

```
$ vault kv metadata delete kv-v2/<key>
```

For the MCG key:

```
$ vault kv metadata delete kv-v2/NOOBAA_ROOT_SECRET_PATH/<key>
```

`<key>` Is the encryption key.

For Example:

```
$ vault kv metadata delete kv-v2/rook-ceph-osd-encryption-key-ocs-deviceset-thin-0-data-0m27q8
```

Example output:

```
Success! Data deleted (if it existed) at: kv-v2/metadata/rook-ceph-osd-encryption-key-ocs-deviceset-thin-0-data-0m27q8
```

iv. Repeat these steps to delete the metadata associated with all the vault keys.

18. To ensure that OpenShift Data Foundation is uninstalled completely, on the OpenShift Container Platform Web Console.

    i. Click *Storage*.
    ii. Verify that OpenShift Data Foundation no longer appears under Storage.

19. If you want to **reuse the disks** on a new ODF installation, or just want to **wipe out** the disks that were used by ceph/osd , please apply steps from this document ODF deployment fails to deploy OSDs due to unclean disks - How to properly erase a disk which was an ODF / Ceph OSD

Removing local storage operator configurations

Use the instructions in this section only if you have deployed OpenShift Data Foundation using local storage devices.

Note: For OpenShift Data Foundation deployments only using localvolume resources, go directly to step 8.

Procedure

1. Identify the `LocalVolumeSet` and the corresponding `StorageClassName` being used by OpenShift Data Foundation.
2. Set the variable SC to the StorageClass providing the `LocalVolumeSet` .

```
$ export SC="<StorageClassName>"
```

3. Delete the `LocalVolumeSet` .

```
$ oc delete localvolumesets.local.storage.openshift.io <name-of-volumeset> -n
openshift-local-storage
```

4. Delete the local storage PVs for the given `StorageClassName` .

```
$ oc get pv | grep $SC | awk '{print $1}'| xargs oc delete pv
```

5. Delete the `StorageClassName` .

```
$ oc delete sc $SC
```

6. Delete the symlinks created by the `LocalVolumeSet` .

```
$ [[ ! -z $SC ]] && for i in $(oc get node -l
cluster.ocs.openshift.io/openshift-storage= -o jsonpath='{
.items[*].metadata.name }'); do oc debug node/${i} -- chroot /host rm -rfv
/mnt/local-storage/${SC}/; done
```

7. Delete the `LocalVolumeDiscovery` .

```
$ oc delete localvolumediscovery.local.storage.openshift.io/auto-discover-devices -n openshift-local-storage
```

8. Remove the LocalVolume resources (if any).

   Use the following steps to remove the LocalVolume resources that were used to provision the PVs in the current or previous OpenShift Data Foundation version. Also, ensure that these resources are not being used by other tenants on the cluster.

   For each of the local volumes, do the following:

   i. Identify the `LocalVolume` and the corresponding `StorageClassName` being used by OpenShift Data Foundation.
   ii. Set the variable LV to the name of the `LocalVolume` and variable SC to the name of the `StorageClass` .
   For example:

```
$ LV=localblock
$ SC=localblock
```

   iii. Delete the local volume resource.

```
$ oc delete localvolume -n openshift-local-storage --wait=true $LV
```

   iv. Delete the remaining PVs and StorageClasses if they exist.

```
$ oc delete pv -l storage.openshift.com/local-volume-owner-name=${LV} --wait --timeout=5m
$ oc delete storageclass $SC --wait --timeout=5m
```

   v. Clean up the artifacts from the storage nodes for that resource.

```
$ [[ ! -z $SC ]] && for i in $(oc get node -l cluster.ocs.openshift.io/openshift-storage= -o jsonpath='{ .items[*].metadata.name }'); do oc debug node/${i} -- chroot /host rm -rfv /mnt/local-storage/${SC}/; done
```

   Example output:

```
Starting pod/node-xxx-debug ...
To use host binaries, run `chroot /host`
removed '/mnt/local-storage/localblock/nvme2n1'
removed directory '/mnt/local-storage/localblock'
Removing debug pod ...
Starting pod/node-yyy-debug ...
To use host binaries, run `chroot /host`
removed '/mnt/local-storage/localblock/nvme2n1'
removed directory '/mnt/local-storage/localblock'
Removing debug pod ...
Starting pod/node-zzz-debug ...
To use host binaries, run `chroot /host`
removed '/mnt/local-storage/localblock/nvme2n1'
removed directory '/mnt/local-storage/localblock'
Removing debug pod ...
```

9. Delete the `openshift-local-storage` namespace and wait till the deletion is complete. You will need to switch to another project if the `openshift-local-storage` namespace is the active project.

   For Example:

   ```
   $ oc project default
   $ oc delete project openshift-local-storage --wait=true --timeout=5m
   ```

   The project is deleted if the following command returns a `NotFound` error.

   ```
   $ oc get project openshift-local-storage
   ```

Removing monitoring stack from OpenShift Data Foundation

Use this section to clean up the monitoring stack from OpenShift Data Foundation.

The Persistent Volume Claims (PVCs) that are created as a part of configuring the monitoring stack are in the `openshift-monitoring` namespace.

Prerequisites

- PVCs are configured to use OpenShift Container Platform monitoring stack.

For more information, see configuring monitoring stack.

Procedure

1. List the pods and PVCs that are currently running in the `openshift-monitoring` namespace.

```
$ oc get pod,pvc -n openshift-monitoring
Example output:
NAME                           READY   STATUS    RESTARTS   AGE
pod/alertmanager-main-0         3/3     Running   0          8d
pod/alertmanager-main-1         3/3     Running   0          8d
pod/alertmanager-main-2         3/3     Running   0          8d
pod/cluster-monitoring-
operator-84457656d-pkrxm        1/1     Running   0          8d
pod/grafana-79ccf6689f-2ll28    2/2     Running   0          8d
pod/kube-state-metrics-
7d86fb966-rvd9w                 3/3     Running   0          8d
pod/node-exporter-25894         2/2     Running   0          8d
pod/node-exporter-4dsd7         2/2     Running   0          8d
pod/node-exporter-6p4zc         2/2     Running   0          8d
pod/node-exporter-jbjvg         2/2     Running   0          8d
pod/node-exporter-jj4t5         2/2     Running   0          6d18h
pod/node-exporter-k856s         2/2     Running   0          6d18h
pod/node-exporter-rf8gn         2/2     Running   0          8d
pod/node-exporter-rmb5m         2/2     Running   0          6d18h
pod/node-exporter-zj7kx         2/2     Running   0          8d
pod/openshift-state-metrics-
59dbd4f654-4clng                3/3     Running   0          8d
pod/prometheus-adapter-
5df5865596-k8dzn                1/1     Running   0          7d23h
pod/prometheus-adapter-
5df5865596-n2gj9                1/1     Running   0          7d23h
pod/prometheus-k8s-0            6/6     Running   1          8d
pod/prometheus-k8s-1            6/6     Running   1          8d
pod/prometheus-operator-
55cfb858c9-c4zd9                1/1     Running   0          6d21h
pod/telemeter-client-
78fc8fc97d-2rgfp                3/3     Running   0          8d

NAME                                                          STATUS
VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS                  AGE
persistentvolumeclaim/my-alertmanager-claim-alertmanager-main-0   Bound
pvc-0d519c4f-15a5-11ea-baa0-026d231574aa   40Gi       RWO              ocs-
storagecluster-ceph-rbd   8d
persistentvolumeclaim/my-alertmanager-claim-alertmanager-main-1   Bound
pvc-0d5a9825-15a5-11ea-baa0-026d231574aa   40Gi       RWO           ocs-
storagecluster-ceph-rbd   8d
persistentvolumeclaim/my-alertmanager-claim-alertmanager-main-2   Bound
pvc-0d6413dc-15a5-11ea-baa0-026d231574aa   40Gi       RWO           ocs-
storagecluster-ceph-rbd   8d
persistentvolumeclaim/my-prometheus-claim-prometheus-k8s-0        Bound
pvc-0b7c19b0-15a5-11ea-baa0-026d231574aa   40Gi       RWO           ocs-
storagecluster-ceph-rbd   8d
persistentvolumeclaim/my-prometheus-claim-prometheus-k8s-1        Bound
pvc-0b8aed3f-15a5-11ea-baa0-026d231574aa   40Gi       RWO           ocs-
storagecluster-ceph-rbd   8d
```

2. Edit the monitoring `configmap` .

```
$ oc -n openshift-monitoring edit configmap cluster-monitoring-config
```

3. Remove any config sections that reference the OpenShift Data Foundation storage classes as shown in the following example and save it.
   Before editing:

```
[...]
apiVersion: v1
data:
 config.yaml: |
   alertmanagerMain:
     volumeClaimTemplate:
       metadata:
         name: my-alertmanager-claim
       spec:
         resources:
           requests:
             storage: 40Gi
         storageClassName: ocs-storagecluster-ceph-rbd
   prometheusK8s:
     volumeClaimTemplate:
       metadata:
         name: my-prometheus-claim
       spec:
         resources:
           requests:
             storage: 40Gi
         storageClassName: ocs-storagecluster-ceph-rbd
kind: ConfigMap
metadata:
 creationTimestamp: "2019-12-02T07:47:29Z"
 name: cluster-monitoring-config
 namespace: openshift-monitoring
 resourceVersion: "22110"
 selfLink: /api/v1/namespaces/openshift-monitoring/configmaps/cluster-
monitoring-config
 uid: fd6d988b-14d7-11ea-84ff-066035b9efa8
[...].
```

   After editing:

```
[...]
apiVersion: v1
data:
 config.yaml: |
kind: ConfigMap
metadata:
 creationTimestamp: "2019-11-21T13:07:05Z"
 name: cluster-monitoring-config
 namespace: openshift-monitoring
 resourceVersion: "404352"
 selfLink: /api/v1/namespaces/openshift-monitoring/configmaps/cluster-
monitoring-config
 uid: d12c796a-0c5f-11ea-9832-063cd735b81c
[...]
```

In this example, `alertmanagerMain` and `prometheusK8s` monitoring components are using the OpenShift Data Foundation PVCs.

4. Delete the relevant PVCs. Make sure you delete all the PVCs that are consuming the storage classes.

```
$ oc delete -n openshift-monitoring pvc <pvc-name> --wait=true
```

`<pvc-name>` Is the name of the PVC

Removing OpenShift Container Platform registry from OpenShift Data Foundation

Use this section to clean up OpenShift Container Platform registry from OpenShift Data Foundation. If you want to configure alternative storage, see Image registry.

The Persistent Volume Claims (PVCs) that are created as a part of configuring the OpenShift Container Platform registry are in the `openshift-image-registry` namespace.

Prerequisites

- The image registry should have been configured to use an OpenShift Data Foundation PVC.

Procedure

1. Edit the `configs.imageregistry.operator.openshift.io` object and remove the content in the **storage** section.

```
$ oc edit configs.imageregistry.operator.openshift.io
```

Before editing:

```
[...]
storage:
   pvc:
       claim: registry-cephfs-rwx-pvc
[...]
```

After editing:

```
[...]
storage:
  emptyDir: {}
[...]
```

In this example, the PVC is called `registry-cephfs-rwx-pvc` , which is now safe to delete.

2. Delete the PVC.

```
$ oc delete pvc <pvc-name> -n openshift-image-registry --wait=true
```

`<pvc-name>` Is the name of the PVC

Removing the cluster logging operator from OpenShift Data Foundation

Use this section to clean up the cluster logging operator from OpenShift Data Foundation.

The Persistent Volume Claims (PVCs) that are created as a part of configuring the cluster logging operator are in the openshift-logging namespace.

Prerequisites

- The cluster logging instance should have been configured to use the OpenShift Data Foundation PVCs.

Procedure

1. Remove the `ClusterLogging` instance in the namespace.

```
$ oc delete clusterlogging instance -n openshift-logging --wait=true --
timeout=5m
```

The PVCs in the openshift-logging namespace are now safe to delete.

2. Delete the PVCs.

```
$ oc delete pvc <pvc-name> -n openshift-logging --wait=true
```

`<pvc-name>` Is the name of the PVC

Removing RADOS Gateway (RGW) component

Use these instructions to disable only the RGW component.

1. Set ignore for RGW in the storage cluster.

```
$ oc patch -n openshift-storage storagecluster ocs-storagecluster \
    --type merge \
    --patch '{"spec": {"managedResources": {"cephObjectStores":
{"reconcileStrategy": "ignore"}}}}'

$ oc patch -n openshift-storage storagecluster ocs-storagecluster \
    --type merge \
    --patch '{"spec": {"managedResources": {"cephObjectStoreUsers":
{"reconcileStrategy": "ignore"}}}}'
```

2. Delete all RGW OBCs

```
$ oc get obc  --all-namespaces --no-headers |grep ceph
#delete all obc from the list above
$ oc delete obc <obc> -n project
```

3. Delete RGW.

```
$ oc delete sc ocs-storagecluster-ceph-rgw
$ oc delete route s3-rgw
$ oc delete service rook-ceph-rgw-ocs-storagecluster-cephobjectstore
$ oc delete CephObjectStore ocs-storagecluster-cephobjectstore
```

Removing Multicloud Object Gateway component

Use these instructions to disable only the MCG component.

1. Set ignore for Multicloud Object Gateway by editing the storage cluster.

```
$ oc edit storagecluster
```

2. Add the two lines marked with arrows.

```
apiVersion: ocs.openshift.io/v1
kind: StorageCluster
metadata:
  name: ocs-storagecluster
  namespace: openshift-storage
spec:
  multiCloudGateway:            #<---
    reconcileStrategy: ignore #<---
```

3. Delete all OBCs.

```
$ oc get obc  --all-namespaces --no-headers |grep noobaa
#delete all OBCs from the list above
$ oc delete obc <obc> -n project
```

4. Allow NooBaa deletion.

```
$ oc patch noobaa noobaa -n openshift-storage --type='merge' -p '{"spec":
{"cleanupPolicy":{"allowNoobaaDeletion":true}}}'
```

5. Delete NooBaa .

```
$ oc delete noobaa noobaa
```

6. Remove the default bucket as described below.

Removing the default bucket created by the Multicloud Object Gateway

The Multicloud Object Gateway (MCG) creates a default bucket in the cloud. You need to remove this default bucket.

NOTE: In case MCG is not deployed, you can skip this section.

Prerequisites

- A running OpenShift Data Foundation Platform.
- Download the MCG command-line interface for easier management:

```
# subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
```

```
# yum install mcg
```

IMPORTANT
Specify the appropriate architecture for enabling the repositories using the subscription manager.

- For IBM Power, use the following command:

```
# subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
```

- For IBM Z infrastructure, use the following command:

```
# subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
```

- Alternatively, you can install the MCG package from the OpenShift Data Foundation RPMs found at the Download RedHat OpenShift Data Foundation page.

NOTE: Choose the correct Product Variant according to your architecture.

Procedure

1. Identify the target bucket that you need to remove:

```
$ noobaa backingstore status noobaa-default-backing-store
```

Example output:

```
[...]
# BackingStore spec:
s3Compatible:
    endpoint: https://**rook-ceph-rgw-ocs-storagecluster-cephobjectstore.openshift-
storage.svc:443**
    secret:
        name: rook-ceph-object-user-ocs-storagecluster-cephobjectstore-noobaa-ceph-
objectstore-user
        namespace: openshift-storage
   signatureVersion: v4
   targetBucket: nb.1648032196977.apps.cluster1.vmware.ocp.team
type: s3-compatible

# Secret data:
[...]
```

In this example, `rook-ceph-rgw-ocs-storagecluster-cephobjectstore.openshift-storage.svc:443` is the endpoint which indicates the service that this backing store is using, for example, rook-ceph, Amazon Web Service (AWS) S3, Azure, and so on. The value in the `targetBucket` field is the name of the bucket that you need to delete.

2. Access the cloud or S3 compatible vendor, and delete the bucket identified in the previous step.

Removing the disaster-protection finalizers in mon and mon endpoints

Starting with 4.12, if you have configured disaster recovery for the clusters, you need to manually remove the 'ceph.rook.io/disaster-protection' finalizer from mon and the mon endpoints:

- Edit `configmap` and `secrets` to remove the `ceph.rook.io/disaster-protection` finalizer from mon and mon endpoints.

```
$ oc edit configmap -n openshift-storage rook-ceph-mon-endpoints
$ oc edit secrets  -n openshift-storage rook-ceph-mon
```

Removing finalizer on the `ocs-client-operator-config` ConfigMap

If OpenShift Data Foundation version 4.16.0 or 4.16.1 is the initial deployment, remove the finalizer on the `configmap` as follows:

```
$ oc patch configmap ocs-client-operator-config -n openshift-storage -p
'{"metadata":{"finalizers":null}}' --type=merge
```

Removing the `csv.odf.openshift.io` mutatingwebhookconfigurations

If you plan on reinstalling ODF at some point, it's necessary to remove the `csv.odf.openshift.io` mutatingwebhookconfigurations CR

```
$  oc delete mutatingwebhookconfigurations csv.odf.openshift.io
mutatingwebhookconfiguration.admissionregistration.k8s.io "csv.odf.openshift.io"
deleted
```

SBR **OCS** **Product(s)** **Red Hat OpenShift Data Foundation** **Article Type** **General**

Was this helpful?

YES NO

Get notified when this content is updated FOLLOW

People who viewed this article also viewed

How to identify OpenShift Data Foundation cluster with internal or

Chapter 4. Uninstalling OpenShift Data Foundation from

Chapter 4. Uninstalling OpenShift Data Foundation from

external or MCG standalone mode

Solution - Sep 16, 2025

external storage system - Red Hat OpenShift Data Foundation 4.18 Deploying OpenShift Data Foundation in external mode

Documentation - Jul 2, 2025

external storage system - Red Hat OpenShift Data Foundation 4.15 Deploying OpenShift Data Foundation in external mode

Documentation - Aug 13, 2025

Case Links (Red Hat Internal)

04367404 - Salesforce / CaseView+ - rhn-support-lsantann

04402290 - Salesforce / CaseView+ - rhn-support-mduasope

04395744 - Salesforce / CaseView+ - rhn-support-rlaberin

04392217 - Salesforce / CaseView+ - rhn-support-rafrojas

04382168 - Salesforce / CaseView+ - rhn-support-rlaberin

04369400 - Salesforce / CaseView+ - rhn-support-snipp

04380543 - Salesforce / CaseView+ - rhn-support-abhishku

04371878 - Salesforce / CaseView+ - rhn-support-snipp

04355572 - Salesforce / CaseView+ - rhn-support-msee1

04354408 - Salesforce / CaseView+ - rhn-support-shivgupt

Show More

Comments

RED HAT

COMMUNITY MEMBER

Add comment

☑ **Send notifications to content followers**

☐ **Mark comment as private**

Submit

RED HAT

COMMUNITY MEMBER

40 Points

Private Comment Mar 7, 2022 8:27 AM

Eran Tamir

need to add a section for deleting the default bucket in the cloud, created by MCG

Reply Privately

RED HAT

NEWBIE

5 Points

Private Comment Apr 27, 2022 4:34 AM

Agil Antony

A new section is added,

Section name: "Removing the default bucket created by the Multcloud Object Gateway"

Location: Right after section, "Removing the cluster logging operator from OpenShift Data Foundation"

Reply Privately

RED HAT

ACTIVE CONTRIB...

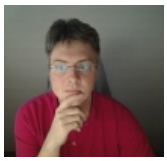
174 Points

Private Comment Oct 28, 2022 3:43 AM

Miguel Duaso

Not sure if this KCS document was updated as per discussion on case 03234405 / Bug 2093424 - [GSS][ODF 4.9] uninstall stuck on step "oc delete -n openshift-storage storagesystem --all --wait=true" ..adding here this for reference. The problem I see is "Determine the required resolution for the failure and implement it." ...as Nitin explained "There can be hundreds of possibilities and we can not cover all of them. I would say from here onwards lets customer figure out whats the issue or let him/her come to us." ...maybe this was address with this section: Chapter 8. Troubleshooting and deleting remaining resources during Uninstall https://access.redhat.com/documentation/en-us/red_hat_openshift_data_foundation/4.10/html-single/troubleshooting_openshift_data_foundation/index#troubleshooting-and-deleting-remaining-resources-during-uninstall_rhodf

Reply Privately



RED HAT

ACTIVE CONTRIB...

183 Points

Private Comment Jan 28, 2023 12:39 PM

Daniel Moessner

I am doing some testing and see the same problem with ODF 4.10 and 4.11: `oc delete -n openshift-storage storagesystem --all --wait=true` lasts for ever and figuring out why is quite tedious. However, as the point is anyway to uninstall, I opened another window and simply did a `oc delete po --all -n openshift-storage` et voila, the first command completes pretty quickly

Reply Privately

RED HAT

PRO

611 Points

Oct 31, 2022 3:11 PM

Nathan Revo RHCSA

When re-installing against a disk that was part of a prior, failed, installation it may be necessary to remove the disk headers/footers laid out by CEPH. Use `sgdisk --zap-all /dev/<disk>` to clean the disk for reuse

Reply

RED HAT

PRO

635 Points

Private Comment May 16, 2023 7:04 PM

Kelson White

These links point back to the same document and need to be updated...:

Delete OpenShift Container Platform monitoring stack PVCs using OpenShift Data Foundation. For more information, see, Removing monitoring stack from OpenShift Data Foundation.

Delete OpenShift Container Platform Registry PVCs using OpenShift Data Foundation. For more information, see, Removing OpenShift Container Platform registry from OpenShift Data Foundation.

Delete OpenShift Container Platform logging PVCs using OpenShift Data Foundation. For more information, see, Removing the cluster logging operator from OpenShift Data Foundation.

Also, why did we remove the uninstall docs from the ODF documentation and put them into a KCS?

Heres the links in the ODF docs:
https://access.redhat.com/documentation/en-us/red_hat_openshift_data_foundation/4.11/html-single/deploying_openshift_data_foundation_using_ibm_z_infrastructure/ir monitoring-stack-from-openshift-data-foundation_ibmz

https://access.redhat.com/documentation/en-us/red_hat_openshift_data_foundation/4.11/html-single/deploying_openshift_data_foundation_using_ibm_z_infrastructure/ir openshift-container-platform-registry-from-openshift-data-foundation_ibmz

https://access.redhat.com/documentation/en-us/red_hat_openshift_data_foundation/4.11/html-single/deploying_openshift_data_foundation_using_ibm_z_infrastructure/ir cluster-logging-operator-from-openshift-data-foundation_ibmz

Reply Privately

May 23, 2023 6:04 AM

J&B

RED HAT

NEWBIE

12 Points

Javier Barea Palma RHCSA RHCE

Trying to uninstall a fresh ODF+LSO and wahtever the annotation combination is used, this command $ oc delete -n openshift-storage storagesystem --all --wait=true stucks at that point.

$ oc describe -n openshift-storage storagecluster ocs-storagecluster .. Events: Type Reason Age From Message ---- ------ ---- ---- ------- Warning UninstallPending 22m controller_storagecluster uninstall: Waiting on NooBaa system noobaa to be deleted Warning UninstallPending 22m controller_storagecluster Uninstall: Waiting for Ceph RGW Route ocs-storagecluster-cephobjectstore to be deleted Warning UninstallPending 22m controller_storagecluster uninstall: Waiting for CephObjectStoreUser ocs-storagecluster-cephobjectstoreuser to be deleted Warning UninstallPending 22m controller_storagecluster uninstall: Waiting for CephObjectStoreUser prometheus-user to be deleted Warning UninstallPending 22m controller_storagecluster uninstall: Waiting for CephObjectStore ocs-storagecluster-cephobjectstore to be deleted Warning UninstallPending 22m controller_storagecluster uninstall: Waiting for CephFileSystem ocs-storagecluster-cephfilesystem to be deleted Warning UninstallPending 22m controller_storagecluster uninstall: Waiting for CephBlockPool ocs-storagecluster-cephblockpool to be deleted Warning UninstallPending 22m controller_storagecluster uninstall: Waiting for CephCluster to be deleted

PVCs deleted (no OBCs) Deleting all pods in order to help, it does NOT. Keeps on stuck

Reply

NEWBIE

7 Points

Aug 1, 2024 5:02 AM

Abdelmageed Abdelsabour

maybe oc patch -n openshift-storage noobaas/noobaa --type=merge -p '{"metadata": {"finalizers":null}}' can help you out

Reply

RED HAT

COMMUNITY MEMBER

81 Points

Private Comment Jun 30, 2023 1:58 PM

Erik Lalancette RHCOE

It seem to miss a step in the section Removing Multicloud Object Gateway component:

Please refer to this BZ: https://bugzilla.redhat.com/show_bug.cgi?id=1981732 Actually that is in fact the desired behaviour. This BZ was created for us to add this **feature**. We wanted to prevent accidental deletions of NooBaa which results in data loss.

If you want to be able to delete it anyway, you will have to set `cleanupPolicy.allowNoobaaDeletion: true` in your NooBaa CR. In order to do so from ansible script, just send a merge patch with previously mentioned attribute set. Once that is in place, noobaa's admission controller will no longer prevent the deletion.

Here is what my customer use and works for them: oc edit noobaa noobaa

replace:

cleanupPolicy: {}

for :

cleanupPolicy:

```
  allowNoobaaDeletion: true
```

Retry the delete :

oc delete noobaa noobaa

noobaa.noobaa.io "noobaa" deleted

Reply Privately

ACTIVE CONTRIB...

164 Points

Jul 26, 2023 1:28 PM

Chet Hosey RHCOE RHCJA

This process doesn't clean up the SCCs created for ODF. Based on my test cluster, those are:

- rook-ceph-csi - https://github.com/rook/rook/blob/master/deploy/examples/operator openshift.yaml
- rook-ceph - https://github.com/rook/rook/blob/master/deploy/examples/operator openshift.yaml
- ocs-metrics-exporter - https://github.com/red-hat-storage/ocs-operator/blob/release-4.13/metrics/deploy/scc.yaml
- noobaa-endpoint - https://github.com/red-hat-storage/noobaa-operator/blob/master/deploy/scc_endpoint.yaml
- noobaa - https://github.com/red-hat-storage/noobaa-operator/blob/master/deploy/scc.yaml
- noobaa-db - https://github.com/noobaa/noobaa-operator/blob/master/deploy/scc_db.yaml

I expect that admins should also run:

```
oc delete scc rook-ceph-csi rook-ceph ocs-metrics-exporter noobaa-en
```

Reply

NEWBIE

10 Points

Sep 8, 2024 3:57 PM

Umesh Sunnapu

I see that clean up jobs are not showing up.

```
oc get pods -n openshift-storage | grep -i cleanup
```

Due to this, there is still content under /var/lib/rook dir

```
To use host binaries, run `chroot /host`
total 4
drwxr-xr-x. 2  167  167  113 Sep  8 19:52 exporter
drwxr-xr-x. 3 root root   18 Sep  8 19:35 mon-b
drwxr-xr-x. 7 root root 4096 Sep  8 19:36 openshift-storage
```

Reply

RED HAT

COMMUNITY MEMBER

22 Points

Mar 5, 2025 3:09 AM

Muhammad Aslam RHCSA RHCE RHCOE

Need to add reconcileStrategy: ignore for CephNFS as well

Reply

Quick Links

Help

Site Info

Related Sites

About

Red Hat Subscription Value

About Red Hat

Red Hat Jobs

About Red Hat
Jobs
Events
Locations
Contact Red Hat
Red Hat Blog
Inclusion at Red Hat
Cool Stuff Store
Red Hat Summit

Copyright © 2026 Red Hat
Privacy statement
Terms of use
All policies and guidelines
Digital accessibility